

A machine-checked absence of real zeros for the primitive even character of conductor 66136

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Abstract

We record a Lean 4 theorem that the Dirichlet L -function attached to the primitive even quadratic character $\eta = \chi_{-66136}$ has no zero on the open real interval $(0, 1)$. The argument is specialised to this single character: positivity of the associated Gaussian theta series on $[1, \infty)$ implies that the completed function is strictly positive for every real $\sigma \in (1/2, 1)$, independently of the sign of the root number; the functional equation reflects the statement to $(0, 1/2)$; and the root number itself is $+1$, obtained by evaluating the theta inversion at the fixed point $x = 1$ rather than by invoking a general theorem on the sign of quadratic Gauss sums. As a separate effective statement, already used in the same development, one has $\|L(1, \eta)\| > 3/2$. Both results are machine-checked against Mathlib and depend only on the standard Lean axioms `propext`, `Classical.choice` and `Quot.sound`. They are independent of Artin's holomorphy conjecture; their intended use in an S_5 -programme is only to remove this quadratic factor as a source of real zeros in a Brauer factorisation.

1 Introduction

Siegel's theorem asserts that a real primitive Dirichlet character has no zero in a neighbourhood of $s = 1$, but the implied constant is ineffective. For a single explicit character the question is finite and, in principle, computational. The present note records a fully formal proof, in Lean 4, that one concrete even primitive character has *no* real zero anywhere in $(0, 1)$, including the centre $s = 1/2$.

Let $D = 66136 = 2^3 \cdot 7 \cdot 1181$ and let η be the primitive even quadratic character of conductor D (in the notation of the accompanying Lean library, `etaChar`). Write $L(s, \eta)$ for the associated Dirichlet L -function and $\Lambda(s, \eta)$ for Mathlib's completed function `DirichletCharacter.comPLETEDLFunction`. The main theorem is the following.

Theorem 1 (Lean: `leta_ne_zero_on_unit_interval_real`). *For every real number σ with $0 < \sigma < 1$,*

$$L(\sigma, \eta) \neq 0.$$

A second, earlier, statement used in the same development is an effective lower bound at the edge of the critical strip.

Theorem 2 (Lean: `Leta_two_norm_ge_half` and the strengthened form $\|L(1, \eta)\| > 3/2$).

$$\|L(1, \eta)\| > \frac{3}{2}.$$

Siegel's theorem would give some (unspecified) $\delta > 0$ with no zero in $(1 - \delta, 1)$; Theorem 2 is a numerical lower bound at $s = 1$ itself, and Theorem 1 eliminates every real point of $(0, 1)$. Neither statement concerns complex zeros in the critical strip, and neither is a case of Artin's conjecture.

The proof does not follow the originally planned route of a truncated approximate functional equation, a 650-term head, and a Lipschitz covering of $[1/2, 71/72]$. That infrastructure was built and remains in the library (`letaAFE_identity`, `norm_letaR_le` with tail bound 10^{-6} for $N \geq 650$). The covering became unnecessary once a structural positivity lemma for the theta series was available.

2 The character

The integer $D = 66136$ arises as the conductor of the quadratic character cutting out the quadratic subfield of a concrete S_5 -extension of $\mathbb{Q}$ (the splitting field of $x^5 - x^4 - 3x^3 - 3x^2 - 2x - 2$, discriminant $264544 = 2^5 \cdot 7 \cdot 1181$). In the Lean development η is constructed as a Mathlib `DirichletCharacter` of modulus D , proved primitive and quadratic, with $\eta(-1) = 1$. The associated Gauss sum is written

$$\tau(\eta) = \sum_{j \bmod D} \eta(j) e^{2\pi i j/D},$$

and Mathlib supplies $\tau(\eta)^2 = D$ together with the identification $\tau(\eta) = \varepsilon\sqrt{D}$ where $\varepsilon = \text{rootNumber}(\eta)$ satisfies $\varepsilon^2 = 1$. The sign of ε is *not* taken from a general evaluation of quadratic Gauss sums—Mathlib does not contain that theorem—and is obtained later from theta positivity (Section 7).

3 Completed function and Mellin split

Mathlib realises the functional equation of an even Dirichlet L -function through the abstract pair `WeakFEPair` instantiated at `hurwitzEvenFEPair`. Specialising to η yields a completed function $\Lambda(s, \eta)$ satisfying

$$\Lambda(s, \eta) = \varepsilon \Lambda(1 - s, \eta)$$

and related to $L(s, \eta)$ by a non-vanishing archimedean factor $\Gamma_{\mathbb{R}}(s)$ for $\text{Re } s > 0$. The precise conductor power in Mathlib’s normalisation is *not* the classical $(D/\pi)^{s/2}\Gamma(s/2)$; it was pinned in Lean by evaluating the termwise identity at $\sigma = 2$, inside the half-plane of absolute convergence, against $L(2, \eta) \Gamma_{\mathbb{R}}(2)$ (`letaAFE_normalisation`).

Write

$$\theta_{\eta}(x) = \sum_{n \in \mathbb{Z}} \eta(n) e^{-\pi n^2 x/D}, \quad x > 0,$$

for Mathlib’s Gaussian theta series attached to η . Poisson summation for the Hurwitz even kernel, twisted by the Gauss-sum shift available for primitive characters, gives the inversion

$$\theta_{\eta}(1/x) = \varepsilon \sqrt{x} \theta_{\eta}(x) \quad (x > 0) \tag{1}$$

(`theta_eta_inversion_rootNumber`). Splitting the Mellin integral of θ_{η} at the self-dual point $x = 1$ then produces

$$\Lambda(\sigma, \eta) = \frac{1}{2} D^{-\sigma/2} \left(T(\sigma/2) + \varepsilon T((1 - \sigma)/2) \right), \tag{2}$$

where

$$T(a) = \int_1^{\infty} x^{a-1} \theta_{\eta}(x) dx.$$

A termwise incomplete-gamma expansion of (2) is also proved (`letaAFE_identity`) but is not required for Theorem 1.

4 Positivity of the theta series

Lemma 3 (Lean: `thetaEtaR_pos`). *For every real $x \geq 1$, $\theta_\eta(x) > 0$.*

Since η is even and $\eta(0) = 0$,

$$\theta_\eta(x) = 2 \sum_{n \geq 1} \eta(n) q^{n^2}, \quad q = e^{-\pi x/D}.$$

Let $A(n) = \sum_{m \leq n} \eta(m)$. A finite kernel check on the explicit character table shows

$$A(n) \geq 0 \quad \text{for all } n \leq 650.$$

(The inequality is false globally: the first negative partial sum occurs near $n = 33069$. The bound 650 is therefore essential and is the same cutoff already used for the incomplete-gamma tail.) Abel summation against the decreasing weights q^{n^2} produces a strictly positive contribution from $n \leq 650$. The tail $n > 650$ is estimated geometrically and shown to be at most one quarter of that head, whence

$$\theta_\eta(x) \geq q - q^4 > 0 \quad (x \geq 1).$$

A quantitative slab $\theta_\eta(x) \geq 1/5$ on $10^4 \leq x \leq 2 \cdot 10^4$ is recorded for explicit Mellin gaps, but strict positivity on $[1, \infty)$ is the structural input.

Corollary 4. *The map $a \mapsto T(a)$ is strictly positive and strictly decreasing on $(0, \infty)$. In particular, if $0 < b < a$ then $T(a) - T(b) > 0$.*

5 Non-vanishing on $(1/2, 1)$

Fix a real σ with $1/2 < \sigma < 1$ and set $a = \sigma/2$, $b = (1 - \sigma)/2$. Then $0 < b < a$, so Corollary 4 gives $0 < T(b) < T(a)$. For either choice $\varepsilon = \pm 1$,

$$T(a) + \varepsilon T(b)$$

is therefore a sum or a difference of two positive numbers of unequal size, hence nonzero (in fact positive after the global sign of (2) is taken into account). The archimedean factor does not vanish on $(0, \infty)$, and one obtains:

Proposition 5 (Lean: `LFunction_etaChar_ne_zero_of_mem`). *If $1/2 < \sigma < 1$, then $L(\sigma, \eta) \neq 0$.*

At the single point $\sigma = 2/3$ the same comparison is made quantitative: $T(1/3) - T(1/6) \geq 3/10$ and $D^{-1/3} \geq 1/41$, so both sign choices for the completed value are bounded below by $3/820$, against an incomplete-gamma tail of 10^{-6} if one prefers that language. The covering argument is not used.

6 Reflection

The functional equation $\Lambda(s) = \varepsilon \Lambda(1 - s)$ together with $\Gamma_{\mathbb{R}}(\sigma) \neq 0$ for $\sigma > 0$ implies that real zeros in $(0, 1)$ are symmetric about $1/2$. Combined with Proposition 5 and with previously established zero-free neighbourhoods of the endpoints

$$(0, 1/72] \cup [71/72, 1]$$

(obtained from a windowed partial-sum estimate against $\|L(1, \eta)\| > 3/2$), one gets:

Proposition 6 (Lean: `leta_ne_zero_off_half`). *If $0 < \sigma < 1$ and $\sigma \neq 1/2$, then $L(\sigma, \eta) \neq 0$.*

The endpoint neighbourhoods are logically redundant once Proposition 5 is reflected, but they were the first effective zero-free intervals obtained in the development and remain in the library.

7 The centre and the root number

At $\sigma = 1/2$ one has $a = b = 1/4$, so (2) collapses to

$$\Lambda(1/2, \eta) = \frac{1}{2} D^{-1/4} (1 + \varepsilon) T(1/4).$$

If $\varepsilon = -1$ this vanishes identically; if $\varepsilon = +1$ it does not. Thus the centre is equivalent to the sign of the root number.

Substitute $x = 1$ in (1):

$$\theta_\eta(1) = \varepsilon \theta_\eta(1).$$

Lemma 3 gives $\theta_\eta(1) > 0$, hence $\varepsilon = +1$ (Lean: `rootNumber_etaChar_eq_one`). Therefore

$$\Lambda(1/2, \eta) = D^{-1/4} T(1/4) > 0,$$

and $L(1/2, \eta) \neq 0$. Together with Proposition 6 this is Theorem 1.

No length- D Gauss sum is expanded. The identity $\tau(\eta)^2 = D$ is used only to know that the root number is ± 1 ; the sign is read off from the already-proved inversion at its fixed point. An independent numerical evaluation $\tau(\eta) = +\sqrt{D}$ is consistent with the theorem but is not part of the formal proof.

8 The effective bound at $s = 1$

Theorem 2 predates the theta argument. It is obtained from the Dirichlet series at $s = 1$,

$$L(1, \eta) = \sum_{n \geq 1} \frac{\eta(n)}{n},$$

by isolating a finite head against an Abel tail controlled by a certified bound $|A(n)| \leq 128$ on the partial sums (valid for all n , not only $n \leq 650$). The resulting lower bound in norm is $\|L(1, \eta)\| > 3/2$. Siegel's theorem yields some ineffective $\delta > 0$ with $L(\sigma, \eta) \neq 0$ on $(1 - \delta, 1)$; the present inequality is a concrete number at a concrete point.

The same partial-sum bound feeds the endpoint zero-free intervals of width $1/72$ used in Section 6.

9 What is not claimed

- $L(s, \eta) \neq 0$ for non-real s in the strip $0 < \operatorname{Re} s < 1$.
- Holomorphy or non-vanishing of any Artin L -function. The character η is Dirichlet; Artin's conjecture is not addressed.
- A general theorem on the sign of quadratic Gauss sums.
- Non-vanishing of the Dedekind zeta function of the degree-6 field appearing in the same S_5 -extension, nor of the five-dimensional Artin factor $L(s, \rho'_5)$.

In the factorisation $\zeta_H = \zeta_F L(\eta) L(\rho'_5)$ attached to that extension, Theorem 1 removes only the quadratic factor as a source of zeros on the real interval $(0, 1)$. The other two factors remain open.

10 Formal status

The theorems live in an isolated Lean library `ArtinS5`, built against a pinned Mathlib checkout. The declarations `leta_ne_zero_on_unit_interval_real`, `leta_ne_zero_at_half`, `rootNumber_etaChar_eq_one`, `thetaEtaR_pos` and `LFunction_etaChar_ne_zero_of_mem` are sorry-free. An isolated axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`. Finite character-table facts (partial sums $A(n)$ for $n \leq 650$, values of η on residue classes) are discharged by kernel evaluation against an explicit table, not by `native_decide` on an unbounded search.

11 Notes on the collaboration

The formal texts were produced by the automated Lean agent Aristotle against a sequence of narrowly scoped instructions. The analytic route—in particular the decision to extract the root-number sign from $\theta_\eta(1)$ rather than to grid the incomplete-gamma head, and the decision not to axiomatise Gauss’s sign theorem—was arrived at jointly in the surrounding analysis. The present note is a record of the theorem that resulted, not a claim that any one author proved it by hand.

References

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